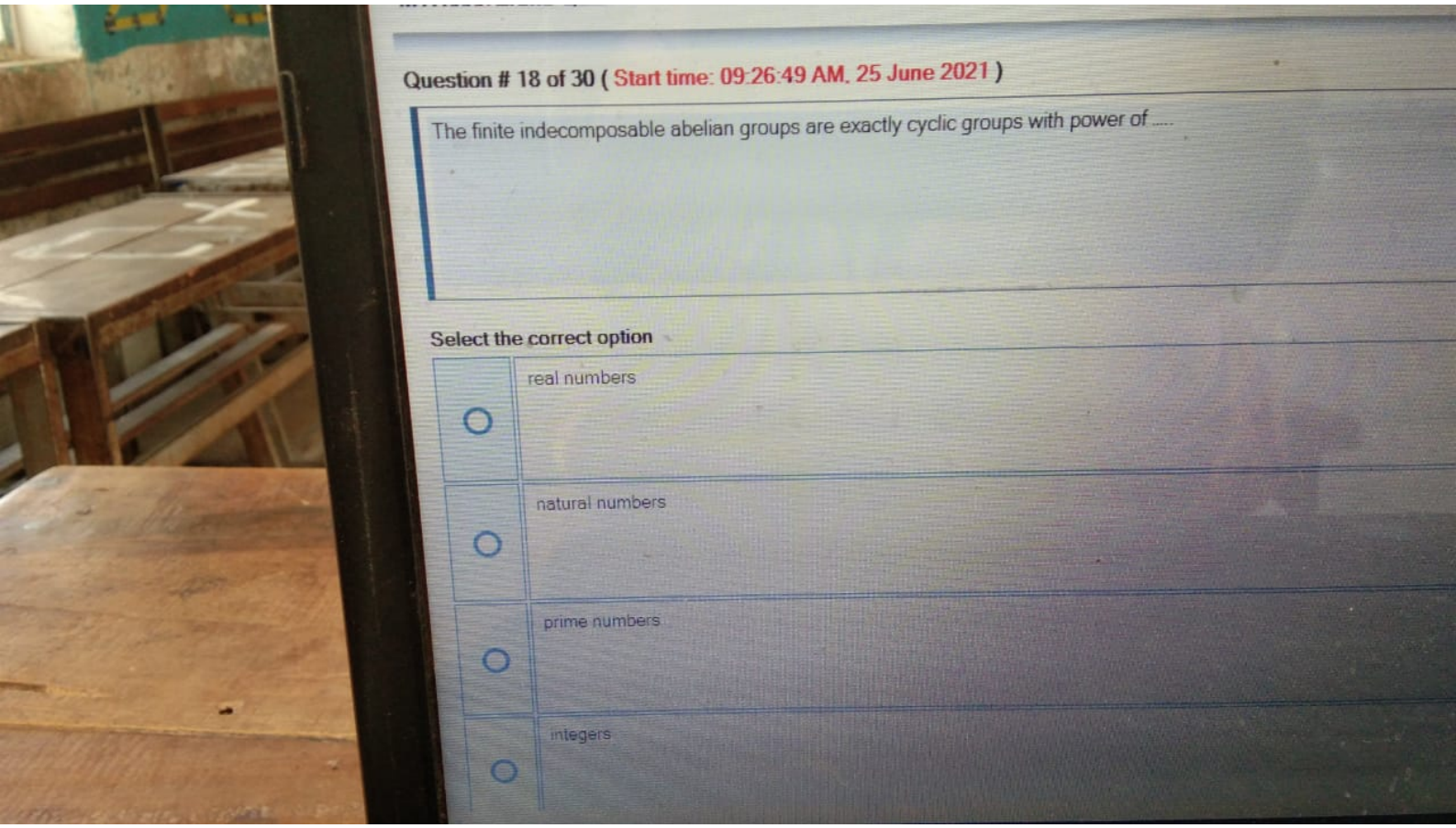


Question # 18 of 30 (Start time: 09:26:49 AM, 25 June 2021)

The finite indecomposable abelian groups are exactly cyclic groups with power of .....

Select the correct option

- real numbers
- natural numbers
- prime numbers
- integers

9:24 95%

Quiz
quiz.vu.edu.pk

stion # 23 of 30 (Start time: 09:24:33 AM, 25 June 2021)

Every element of a group has ---------- inverse(s).

ect the correct option

- 4
- 1
- 3
- 2

Click to Save Answer & Move to Ne

9:22 0.40 KB/s 90

Quiz
quiz.vu.edu.pk

Question # 18 of 30 (Start time: 09:25:02 AM, 25 June 2021) Total

Every group of prime order is ....

Select the correct option

- none of above
- cyclic group
- non-cyclic group
- non-abelian group

Click to Save Answer & Move to Next Question

9:21 4.20 KB/s 90

Quiz
quiz.vu.edu.pk

Question # 17 of 30 (Start time: 09:23:18 AM, 25 June 2021)

Total Marks:

Let f : G → H be group morphism, the kernel of function f is defined as

Select the correct option

- ker f={f(h) = e', h in H}
- ker f={f(g) = e, g in G}
- ker f={f(h) = e, h in H}
- ker f={f(g) = e', g in G}

Click to Save Answer & Move to Next Question

Jazz
ZONG
95% 9:25 AM
Quiz
quiz.vu.edu.pk

MC190401108: HAFIZ MUHAMMAD RIZWAN

Time Left 86 sec(s)

MTH633:Grand Quiz

Quiz Start Time: 09:11 AM, 25 June 2021

Question # 16 of 30 (Start time: 09:25:14 AM, 25 June 2021)

Total Marks: 1

How many tests are there to show a subset of a group to be a subgroup?

Select the correct option

- 3
- 2
- 1
- 4

Click to Save Answer & Move to Next Question

9:22 95%

Quiz
quiz.vu.edu.pk

Question # 21 of 30 (Start time: 09:22:50 AM, 25 June 2021)

In the group {1, -1, i, -i}, the order of -1 is -------

Select the correct option

- 2
- 1
- 3
- 4

Click to Save Answer & Move to Next

Jazz ZONG 95% 9:24 AM

Quiz
quiz.vu.edu.pk

MC190401108: HAFIZ MUHAMMAD RIZWAN

Time Left 88 sec(s)

MTH633:Grand Quiz

Quiz Start Time: 09:11 AM, 25 June 2021

Question # 15 of 30 (Start time: 09:24:09 AM, 25 June 2021)

Total Marks: 1

Let h be a homomorphism from

$$G$$

into

$$G_1$$

. Let K be kernel of h then

Select the correct option

Reload Math Equations

- $aK = h^{-1}[\{ h^{-1}(a)\}]$
- $aK = h^{-1}[\{ h(a)\}]$
- $aK = h[\{ h^{-1}(a)\}]$
- $aK = h[\{ h(a)\}]$

Click to Save Answer & Move to Next Question

9:23 95%

Quiz
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MTH633:Grand Quiz

Question # 22 of 30 (Start time: 09:23:39 AM, 25 June 2021)

Alternating group A_4 which has 12 elements has no subgroup of order

Select the correct option

- 4
- 3
- all of above
- 6

Click

Jazz
ZONG
95% 9:23 AM
Quiz
quiz.vu.edu.pk

MC190401108: HAFIZ MUHAMMAD RIZWAN

Time Left 88 sec(s)

MTH633:Grand Quiz

Quiz Start Time: 09:11 AM, 25 June 2021

Question # 14 of 30 (Start time: 09:23:11 AM, 25 June 2021)

Total Marks: 1

Every cyclic group of prime order is .....to

Z_p

Select the correct option

Reload Math Equations

- homomorphic
- isomorphic
- automorphic
- endomorphic

Click to Save Answer & Move to Next Question

Quiz
quiz.vu.edu.pk

MC190401108: HAFIZ MUHAMMAD RIZWAN

Time Left 88 sec(s)

MTH633:Grand Quiz

Quiz Start Time: 09:11 AM, 25 June 2021

Question # 13 of 30 (Start time: 09:22:16 AM, 25 June 2021)

Total Marks: 1

The parity of a permutation can be both even and odd.

Select the correct option

- True
- False

Click to Save Answer & Move to Next Question

4G 9:20 21.8 KB/s 90

Quiz
quiz.vu.edu.pk

MTH633:Grand Quiz | Quiz Start Time: 09:09 AM, 25 J

Question # 16 of 30 (Start time: 09:22:15 AM, 25 June 2021) | Tota

Let H be a subgroup of G and a in G. If

$$a \in H$$

then

Select the correct option | Reload Math E

- aH is less than G
- aH is greater than equal to G
- aH is less than equal to G
- aH is greater than G

Click to Save Answer & Move to Next Questio

4G 9:19 28.3 KB/s 90

Quiz
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MTH633:Grand Quiz

Quiz Start Time: 09:09 AM, 25 June

Question # 14 of 30 (Start time: 09:21:14 AM, 25 June 2021)

Total M

Can we solve $10 + x = 3$ in Z?

Select the correct option

Reload Math Equa

- Yes
- No

Click to Save Answer & Move to Next Question

9:17 24.5 KB/s 90

Quiz
quiz.vu.edu.pk

The finite indecomposable abelian groups are exactly cyclic groups with power of .....

Select the correct option

- prime numbers
- natural numbers
- real numbers
- integers

Click to Save Answer & Move to Next Question

Jazz
ZONG
96% 9:21 AM
Quiz
quiz.vu.edu.pk

MC190401108: HAFIZ MUHAMMAD RIZWAN

Time Left 88 sec(s)

MTH633:Grand Quiz

Quiz Start Time: 09:11 AM, 25 June 2021

Question # 12 of 30 (Start time: 09:21:19 AM, 25 June 2021)

Total Marks: 1

The group ({1, -1}, .) is subgroup of the group (Z, +).

Select the correct option

- False
- True

Click to Save Answer & Move to Next Question

9:21 95%

Quiz
quiz.vu.edu.pk

uestion # 20 of 30 (Start time: 09:21:32 AM, 25 June 2021)

$$\frac{1}{(a + bi)} = \frac{a}{(a^2 + b^2)} - \frac{b\,i}{(a^2 + b^2)}$$

elect the correct option

- False
- True

Click to Save Answer & Move

Jazz
ZONG
96% 9:19 AM
Quiz
quiz.vu.edu.pk

MC190401108: HAFIZ MUHAMMAD RIZWAN

Time Left 88 sec(s)

MTH633:Grand Quiz

Quiz Start Time: 09:11 AM, 25 June 2021

Question # 11 of 30 (Start time: 09:19:55 AM, 25 June 2021)

Total Marks: 1

The group (Q, +) is an infinite group, where Q is the set of rational numbers.

Select the correct option

- False
- True

Click to Save Answer & Move to Next Question

9:20 96%

Quiz
quiz.vu.edu.pk

Question # 18 of 30 (Start time: 09:19:59 AM, 25 June 2021)

$\langle Z, + \rangle$ is the subgroup of $\langle R, \cdot \rangle$.

Select the correct option

- True
- False

Click to Save Answ

9:20 96%

Quiz
quiz.vu.edu.pk

A cycle of length _____ is called a transposition.

Select the correct option

- 3
- 2
- 1
- 4

Click to Save Answer & Move to Next Questi

4G 9:16 1.10 KB/s 90

Quiz
quiz.vu.edu.pk

MTH633:Grand Quiz

Quiz Start Time: 09:09 AM, 25 June 202

Question # 12 of 30 (Start time: 09:18:27 AM, 25 June 2021)

Total Marks:

The group Z4 under addition is abelian because it is _____.

Select the correct option

- infinite
- finite
- non-cyclic group
- cyclic group

Click to Save Answer & Move to Next Question

Jazz ZONG 96% 9:19 AM

Quiz
quiz.vu.edu.pk

MC190401108: HAFIZ MUHAMMAD RIZWAN

Time Left 88 sec(s)

MTH633:Grand Quiz

Quiz Start Time: 09:11 AM, 25 June 2021

Question # 10 of 30 (Start time: 09:19:18 AM, 25 June 2021)

Total Marks: 1

For each positive integer n, let Un be the multiplicative group of the nth roots of unity in C. Then the elements of Un can be represented geometrically with equally spaced points on

Select the correct option

- a square
- an ellipse with centre origin
- a rectangle
- a circle with centre origin

Click to Save Answer & Move to Next Question

9:19 96%

Quiz
quiz.vu.edu.pk

Question # 17 of 30 (Start time: 09:19:16 AM, 25 June 2021)

The group {1, -1, i, -i} is a finite group.

Select the correct option

- False
- True

Click to Save Answer &

4G 9:14 0.60 KB/s 91

Quiz
quiz.vu.edu.pk

Question # 10 of 30 (Start time: 09:16:53 AM, 25 June 2021)

If m and n are relatively prime, then cyclic group

$$Z_m \times Z_n$$

is isomorphic to

Select the correct option

Reloc

- Z_{m-n}
- $Z_{m/n}$
- Z_{m+n}
- Z_{mn}

Click to Save Answer & Move to Ne

Jazz ZONG 97% 9:17 AM

Quiz
quiz.vu.edu.pk

MC190401108: HAFIZ MUHAMMAD RIZWAN

Time Left 87 sec(s)

MTH633:Grand Quiz

Quiz Start Time: 09:11 AM, 25 June 2021

Question # 8 of 30 (Start time: 09:17:28 AM, 25 June 2021)

Total Marks: 1

Suppose G = < a > with | a | = 5 and e is identity. Which of the following is true?

Select the correct option

- a^10 = e
- a^13 = e
- a^12 = e
- a^11 = e

Click to Save Answer & Move to Next Question

9:17 96%

Quiz
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Question # 15 of 30 (Start time: 09:16:58 AM, 25 June 2021)

Show that $H = \left\{ \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \mid n \in Z \right\}$ is the subgroup of G = GL (2, R) under multiplication.

Select the correct option

- True
- False

Click to Save Answer & Move to

4G 9:15 20.2 KB/s 91

Quiz
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uestion # 11 of 30 (Start time: 09:17:35 AM, 25 June 2021)
Total Marks:

The set of real numbers R is the union of ----------

elect the correct option

Reload Math Equations

- Z and Q'
- $N \text{and} W$
- Q and Q'
- Q and W

Click to Save Answer & Move to Next Question

9:18 96%

Quiz
quiz.vu.edu.pk

Question # 16 of 30 (Start time: 09:18:05 AM, 25 June 2021)

The group $\{1, -1, i, -i\}$ is not a subgroup of itself.

Select the correct option

- True
- False

Click to

4G 9:14 21.8 KB/s 91

Quiz
quiz.vu.edu.pk

Question # 9 of 30 (Start time: 09:16:08 AM, 25 June 2021) Total

Let

$$\phi : G \to H$$

be a homomorphism, then $\phi(a^{-1}) = \phi(a)^{-1}$ for $a \in G$.

Select the correct option

Reload Math Eq

- False
- True

Click to Save Answer & Move to Next Question

Jazz ZONG 97% 9:16 AM

Quiz
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MC190401108: HAFIZ MUHAMMAD RIZWAN

Time Left 87 sec(s)

MTH633:Grand Quiz

Quiz Start Time: 09:11 AM, 25 June 2021

Question # 7 of 30 (Start time: 09:16:25 AM, 25 June 2021)

Total Marks: 1

Let G be an abelian group. A direct product of G is

Select the correct option

- non-cyclic
- abelian
- cyclic
- non-abelian

Click to Save Answer & Move to Next Question

Jazz ZONG 97% 9:18 AM

Quiz
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MC190401108: HAFIZ MUHAMMAD RIZWAN

Time Left 88 sec(s)

MTH633:Grand Quiz

Quiz Start Time: 09:11 AM, 25 June 2021

Question # 9 of 30 (Start time: 09:18:19 AM, 25 June 2021)

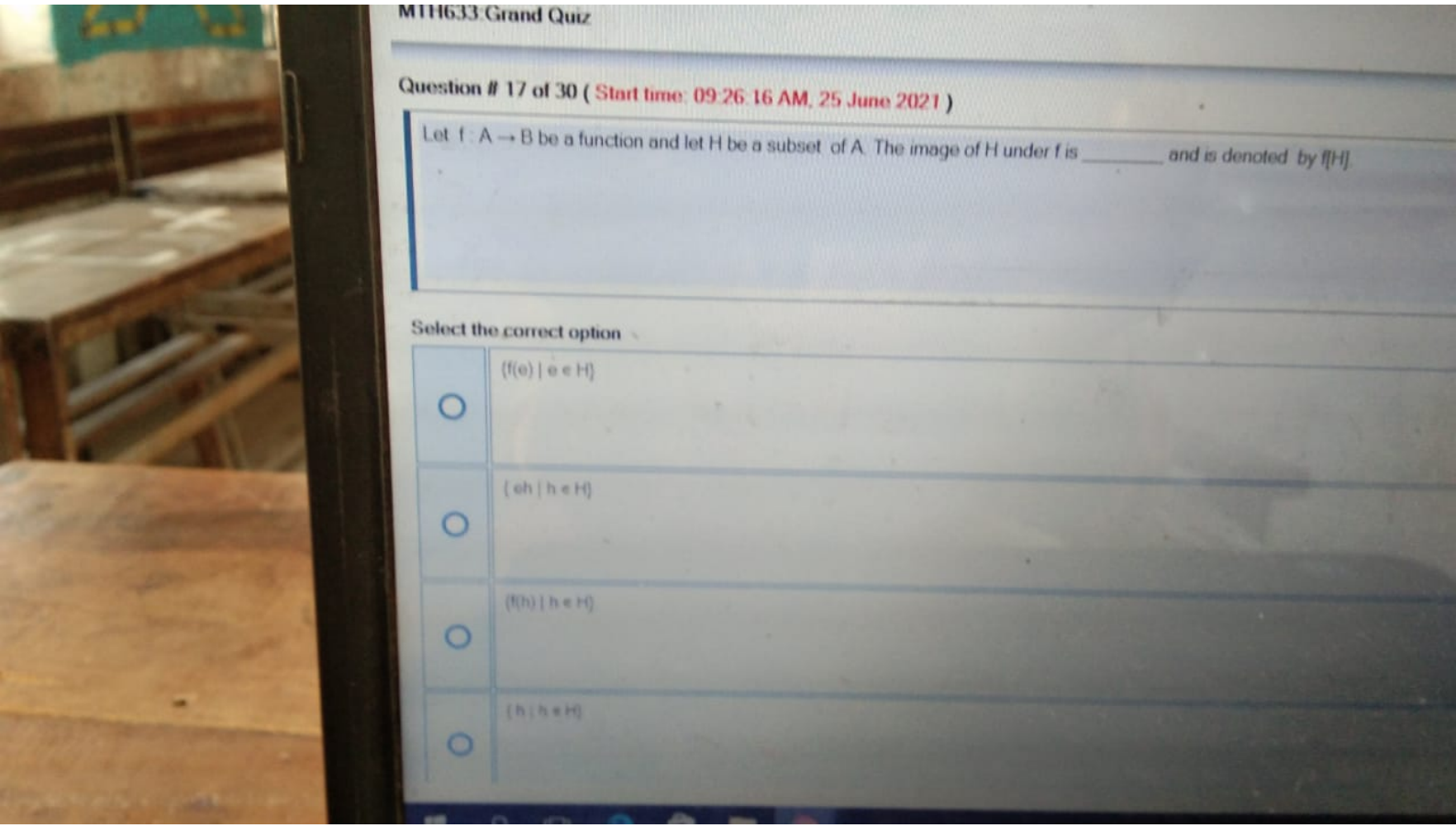
Total Marks: 1

For a cyclic group G = < a >, if |a| = n, then a^i = a^j iff n divides _____

Select the correct option

- i
- i + j
- j
- i - j

Click to Save Answer & Move to Next Question


MTH633 Grand Quiz
Question # 17 of 30 (Start time: 09:26:16 AM, 25 June 2021)
Let f : A → B be a function and let H be a subset of A. The image of H under f is _____ and is denoted by f[H].
Select the correct option
{f(e) | e ∈ H}
{eh | h ∈ H}
{f(h) | h ∈ H}
{h | h ∈ H}

9:15 96%

Quiz
quiz.vu.edu.pk

uestion # 14 of 30 (Start time: 09:15:54 AM, 25 June 2021) Tot

The order of the group {e} is ----------

elect the correct option

- 1
- 0
- 2
- 3

Click to Save Answer & Move to Next Quest

9:15 96%

Quiz
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MTH633:Grand Quiz

Quiz Start Time: 09:05 AM, 25 June 202

Question # 13 of 30 (Start time: 09:14:57 AM, 25 June 2021)

Total Marks:

For each positive integer n, the multiplicative group Un of the nth roots of unity in C is _____

Select the correct option

- cyclic
- dihedral
- non-cyclic
- non-abelian

Click to Save Answer & Move to Next Question

9:13 10.1 KB/s 91

Quiz
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Question # 7 of 30 (Start time: 09:15:13 AM, 25 June 2021) Total Marks

D4 is the group of symmetries of the _____.

Select the correct option

- rhombus
- trapezium
- rectangle
- square

Click to Save Answer & Move to Next Question

Jazz ZONG 98% 9:15 AM

Quiz
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MC190401108: HAFIZ MUHAMMAD RIZWAN

Time Left 88 sec(s)

MTH633:Grand Quiz

Quiz Start Time: 09:11 AM, 25 June 2021

Question # 6 of 30 (Start time: 09:15:31 AM, 25 June 2021)

Total Marks: 1

If G is finite group and H is subgroup of G then |H| divides |G|.

Select the correct option

- Inversion Theorem
- Cayley Theorem
- Sylow Theorem
- Lagrange Theorem

Click to Save Answer & Move to Next Question

9:14 97%

Quiz
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If f : G →H be group homomorphism. If function f is ..... ther ker = {e}

Select the correct option

- sujective
- bijective
- none of above
- injective

Click to Save Answer & Move to Next Quest

Jazz
ZONG
98% 9:14 AM

Quiz
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MC190401108: HAFIZ MUHAMMAD RIZWAN

Time Left 88 sec(s)

MTH633:Grand Quiz

Quiz Start Time: 09:11 AM, 25 June 2021

Question # 4 of 30 (Start time: 09:14:03 AM, 25 June 2021)

Total Marks: 1

A group can have more than one identity.

Select the correct option

- True
- False

Click to Save Answer & Move to Next Question

Jazz
ZONG
98% 9:14 AM
Quiz
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MC190401108: HAFIZ MUHAMMAD RIZWAN

Time Left 88 sec(s)

MTH633:Grand Quiz

Quiz Start Time: 09:11 AM, 25 June 2021

Question # 5 of 30 (Start time: 09:14:50 AM, 25 June 2021)

Total Marks: 1

_____ is also called the octic group.

Select the correct option

- D6
- D3
- D5
- D4

Click to Save Answer & Move to Next Question

4G 9:11 3.20 KB/s 91

Quiz
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6 of 30 (Start time: 09:13:40 AM, 25 June 2021) Total Marks: 1

m divides the order of a finite abelian group G en G has a .... of order m.

e correct option

- subgroup
- cyclic group
- klein 4-group
- subgroup

Click to Save Answer & Move to Next Question

9:13 97%

Quiz
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For a cyclic group G = < a >, if |a| = n, then < a^k > = _____

Select the correct option

- < a ^ gcd(n, k-3) >
- < a ^ gcd(n, k-2) >
- < a ^ gcd(n, k) >
- < a ^ gcd(n, k-1) >

Click to Save Answer & Move to Next Ques

Jazz
ZONG
4G 3G 98% 9:13 AM
Quiz
quiz.vu.edu.pk

MC190401108: HAFIZ MUHAMMAD RIZWAN

Time Left 88 sec(s)

MTH633:Grand Quiz

Quiz Start Time: 09:11 AM, 25 June 2021

Question # 3 of 30 (Start time: 09:13:21 AM, 25 June 2021)

Total Marks: 1

Every group of prime order is ....

Select the correct option

- none of above
- cyclic group
- non-cyclic group
- non-abelian group

Click to Save Answer & Move to Next Question

4G 9:10 0.40 KB/s 91

Quiz
quiz.vu.edu.pk

Question # 5 of 30 (Start time: 09:12:53 AM, 25 June 2021)

The binary structures $\langle Q, +\rangle$ and $\langle R, +\rangle$ are not isomorphic.

Select the correct option

- False
- True

Click to Save Answer &

9:12 97%

Quiz
quiz.vu.edu.pk

Suppose G = < a > with | a | = 7 and e is identity. Which of the following is true?

elect the correct option

- a^14 = e
- a^15 = e
- a^12 = e
- a^13 = e

Click to Save Answer & Move to Next Question

9:12 97%

Quiz
quiz.vu.edu.pk

Question # 9 of 30 (Start time: 09:12:06 AM, 25 June 2021)

Which of the following is NOT a rational number?

Select the correct option

○	5
○	0
○	$\frac{22}{7}$
○	π

Jazz
ZONG
4G 3G 98% 9:12 AM
Quiz
quiz.vu.edu.pk

MC190401108: HAFIZ MUHAMMAD RIZWAN

Time Left 87 sec(s)

MTH633:Grand Quiz

Quiz Start Time: 09:11 AM, 25 June 2021

Question # 2 of 30 (Start time: 09:12:31 AM, 25 June 2021)

Total Marks: 1

The order of group

$$Z_{12} x\ Z_{60} x\ Z_{24}$$

is

Select the correct option

Reload Math Equations

- 12
- 60
- 24
- 8

Click to Save Answer & Move to Next Question

9:10 1.20 KB/s 91

Quiz
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MTH633:Grand Quiz

Quiz Start Time: 09:09 AM, 25 Jur

Question # 4 of 30 (Start time: 09:12:00 AM, 25 June 2021)

Total M

Let G and G' be groups and let $\phi : G \rightarrow G'$ be a one-to-one function such that _____, The $\phi[G]$ is a subgroup of G' and ϕ provides an isomorphism of G with $\phi[G]$

Select the correct option

- $\phi(xy) = \phi(x) - \phi(y)$ for all $x, y \in G$
- $\phi(xy) = \phi(x) / \phi(y)$ for all $x, y \in G$
- $\phi(xy) = \phi(x)\,\phi(y)$ for all $x, y \in G$
- $\phi(xy) = \phi(x) + \phi(y)$ for all $x, y \in G$

Click to Save Answer & Move to Next Question

Jazz ZONG 98% 9:11 AM

Quiz
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MC190401108: HAFIZ MUHAMMAD RIZWAN

Time Left 86 sec(s)

MTH633:Grand Quiz

Quiz Start Time: 09:11 AM, 25 June 2021

Question # 1 of 30 (Start time: 09:11:46 AM, 25 June 2021)

Total Marks: 1

Let G be a finite cyclic group generated by its element a, then _____

Select the correct option

- | G | > | a |
- | G | = | a |
- | G | < | a |
- None of these

Click to Save Answer & Move to Next Question

9:11 97%

Quiz
quiz.vu.edu.pk

MTH633:Grand Quiz

Question # 8 of 30 (Start time: 09:11:15 AM, 25 June 2021)

Let H be a subgroup of G and a in G. If

$$a \in H$$

then

Select the correct option

○	aH is less than equal to G
○	aH is greater than G
○	aH is greater than equal to G
○	aH is less than G

9:09 27.1 KB/s 91

Quiz
quiz.vu.edu.pk

Question # 2 of 30 (Start time: 09:11:04 AM, 25 June 2021)

A subset H of a group G under addition is a subgroup of G if and only if whenever $a, b \in H$, then $a - b \in H$

Select the correct option

- True
- False

Click to Save Answer & M

9:09 97%

Quiz
quiz.vu.edu.pk

Question # 6 of 30 (Start time: 09:09:48 AM, 25 June 2021)

The order of the group {1, -1} is ----------

Select the correct option

- 2
- 4
- 1
- 3

Click to Save Answer & Mov

9:09 97%

Quiz
quiz.vu.edu.pk

Suppose G = < a > with |a| = 30. Then |a^21| = _____

elect the correct option

- 11
- 10
- 12
- 13

Click to Save Answer & Move to Next Q

9:08 15.6 KB/s 92

Quiz
quiz.vu.edu.pk

Question # 1 of 30 (Start time: 09:09:44 AM, 25 June 2021)

Let G and G' be groups and let $\phi : G \rightarrow G'$ be one-to-one function such that $\phi(xy) = \phi(x)$ for all $x, y \in G.$, then _____

Select the correct option

- $\phi[G]$ is equal to G'
- $\phi[G]$ is a subgroup of G'
- None of these
- G' is a subgroup of $\phi[G]$

Click to Save Answer & Move to Next

9:25 95%

Quiz
quiz.vu.edu.pk

Question # 24 of 30 (Start time: 09:25:37 AM, 25 June 2021)

The order of group

$$Z_{12} x\ Z_{60} x\ Z_{24}$$

is

Select the correct option

- 8
- 60
- 24
- 12

4G 9:23 7.10 KB/s 90

Quiz
quiz.vu.edu.pk

Question # 19 of 30 (Start time: 09:25:52 AM, 25 June 2021) Total Marks:

Let G be an abelian group. A direct product of G is

Select the correct option

- cyclic
- non-cyclic
- non-abelian
- abelian

Click to Save Answer & Move to Next Question

Question # 16 of 30 (Start time: 09:25:15 AM, 25 June 2021)

Let f:G→H be group morphism.

and

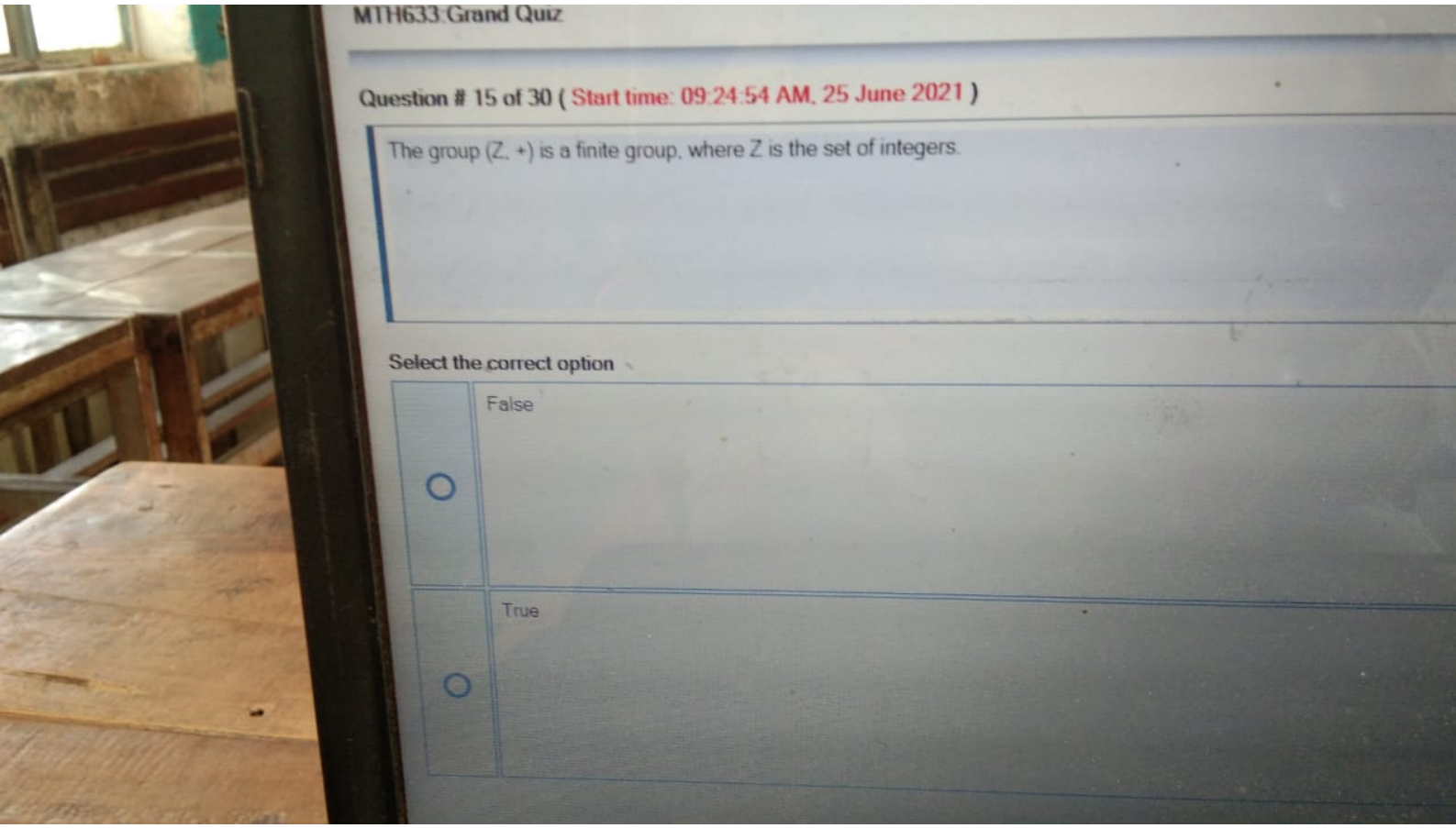
e_G

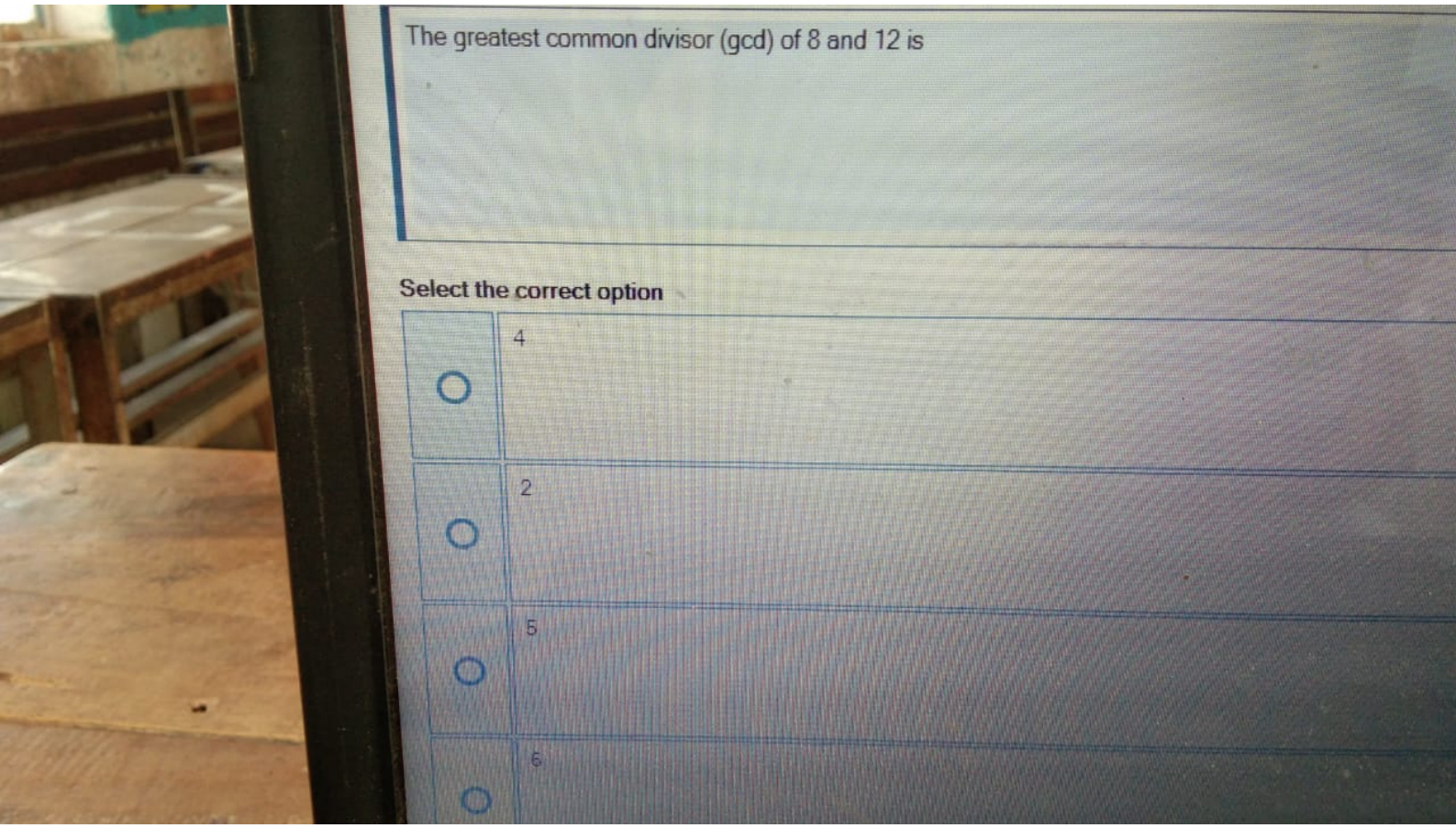
e_H

be identities of G and H respectively then

Select the correct option

- $f(e_G) = e_G$
- $f(e_H) = e_G$
- $f(e_G) = e_H$
- $f(e_H) = e_H$


MTH633 Grand Quiz
Question # 15 of 30 (Start time: 09:24:54 AM, 25 June 2021)
The group (Z, +) is a finite group, where Z is the set of integers.
Select the correct option
False
True


The greatest common divisor (gcd) of 8 and 12 is
Select the correct option
4
2
5
6

Which of the following is true?

Select the correct option

- $D4 \geq S4$
- $D4 = S4$
- $D4 > S4$
- $D4 \leq S4$

Question # 12 of 30 (Start time: 09:23:36 AM, 25 June 2021)

Let S_n be symmeric group on n elements. Let function f from S_n to Z_2 be homomorphism then 1. f is 0 if it is ... permutation 2. f is 1 if it is ... permutation

Select the correct option

- even , odd
- even , even
- odd , even
- odd , odd

Question # 11 of 30 (Start time: 09:23:03 AM, 25 June 2021)

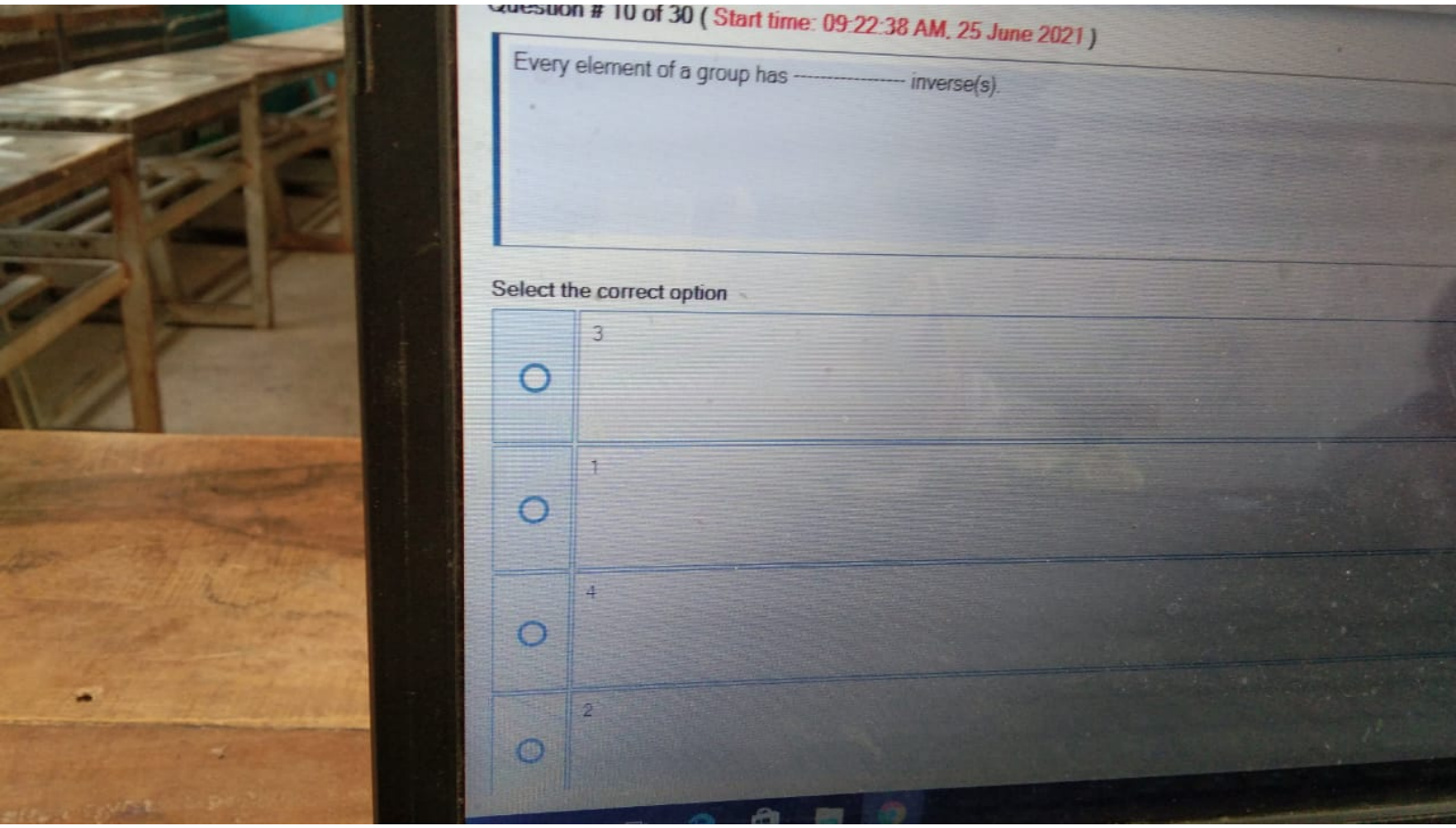
Total number of element in group

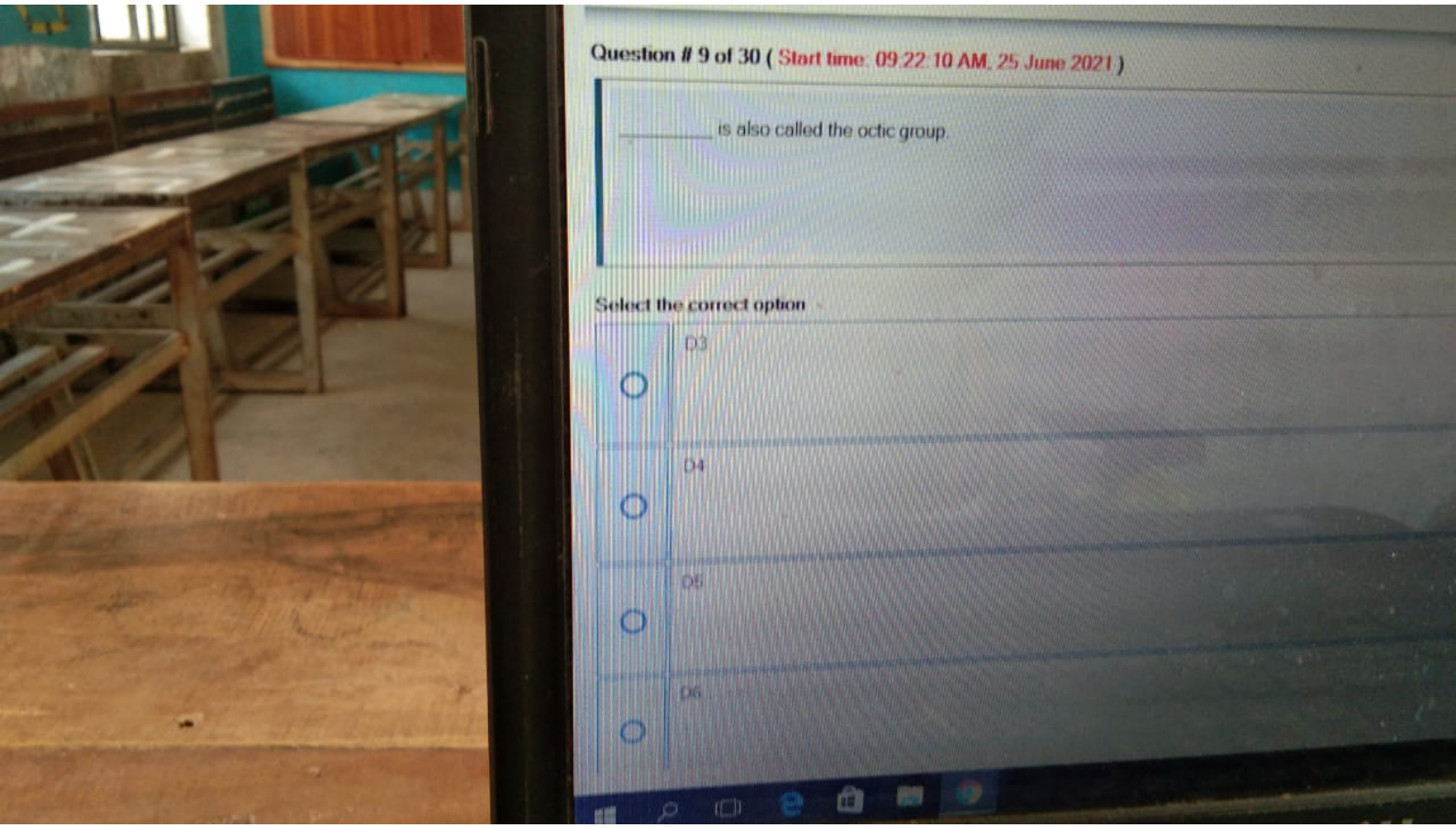
$Z_2 \times Z_3$

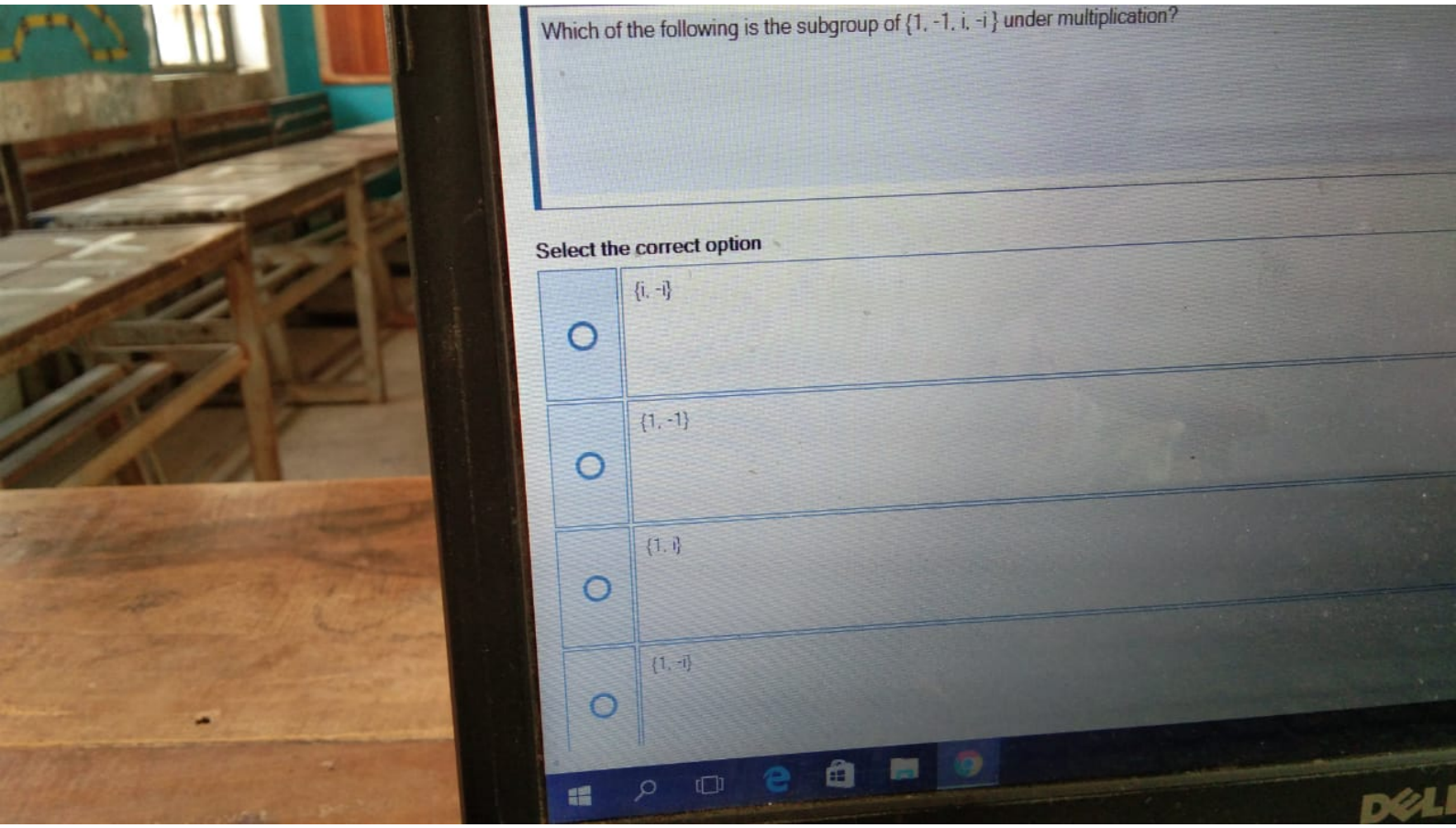
is

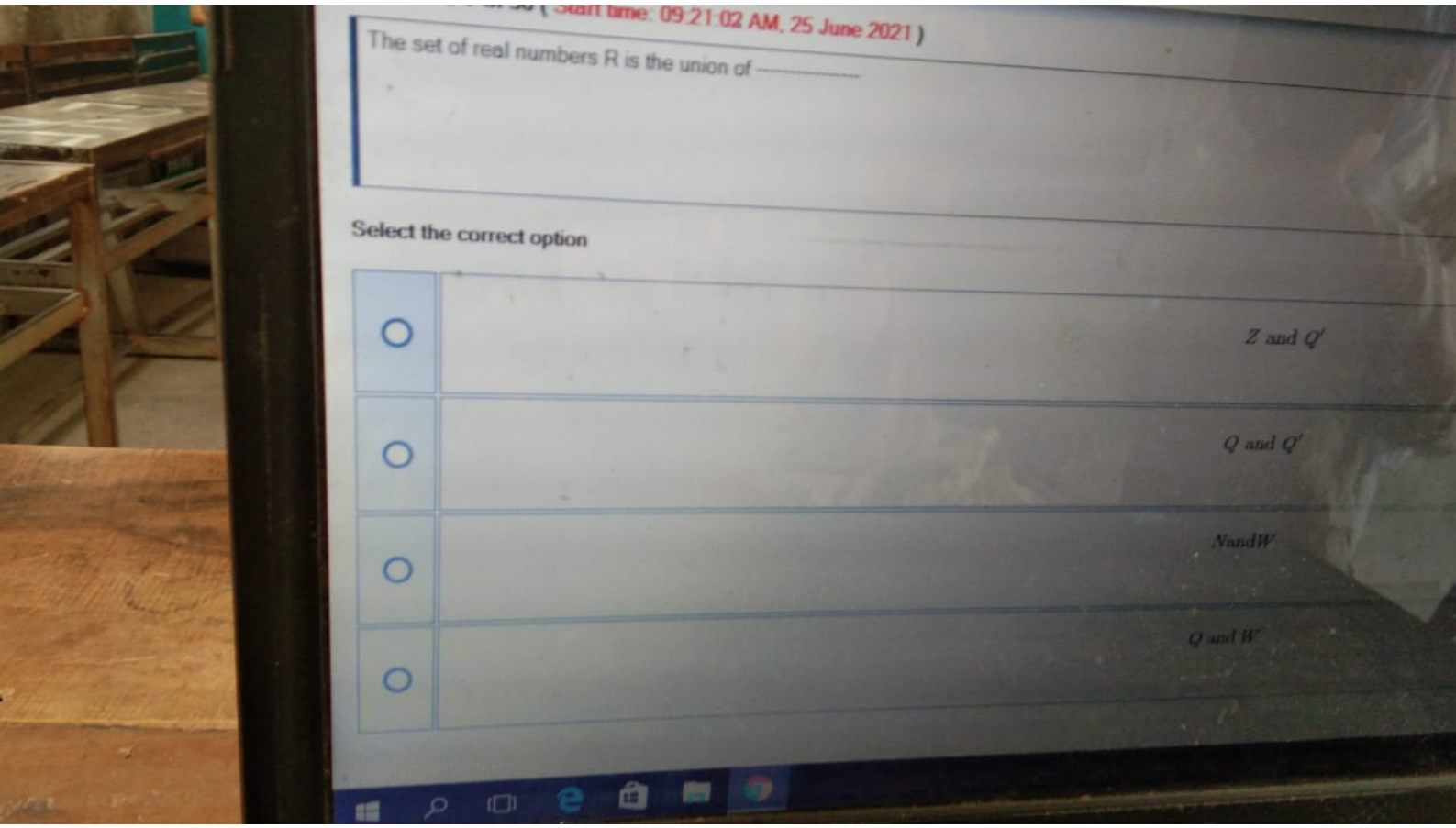
Select the correct option

- 2
- 6
- 5
- 3


Question # 10 of 30 (Start time: 09:22:38 AM, 25 June 2021)
Every element of a group has ---------- inverse(s).
Select the correct option
3
1
4
2


Question # 9 of 30 (Start time: 09:22:10 AM, 25 June 2021)
_____ is also called the octic group.
Select the correct option
D3
D4
D5
D6


Which of the following is the subgroup of {1, -1, i, -i} under multiplication?
Select the correct option
{i, -i}
{1, -1}
{1, i}
{1, -i}


Start time: 09:21:02 AM, 25 June 2021)
The set of real numbers R is the union of ----------
Select the correct option
Z and Q'
Q and Q'
NandW
Q and W